

Dual-LSTM-Based Volatility Prediction

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Abstract—This paper studies short-term stock volatility prediction using price data and financial news sentiment. We compare three LSTM-based models: a price-only model using OHLCV features, a sentiment-only model using daily sentiment scores extracted with FinBERT, and a combined model that fuses price and sentiment embeddings through a fusion DNN. All models are evaluated on the same time-based train-test split using MSE, RMSE, MAE, and MAPE. Experimental results show that the combined model achieves the lowest prediction error, outperforming both price-only and sentiment-only models. The sentiment-only model performs worst, partly due to treating days without news as neutral.

Index Terms—Volatility prediction, LSTM, sentiment analysis, FinBERT, financial time series.

I. INTRODUCTION

Volatility is a practical measure of market risk and is often more useful than predicting future prices, which can be noisy and unstable. Accurate short-term volatility forecasting supports risk-aware decision making in applications such as portfolio management and trading.

This project investigates whether combining historical price patterns with financial news sentiment improves volatility prediction. We build and compare three LSTM-based models: a price-only model using daily OHLCV features, a sentiment-only model using daily sentiment scores extracted from news with FinBERT, and a combined model that fuses price and sentiment embeddings through a small neural network.

Using a time-based train-test split and evaluating with MSE, RMSE, MAE, and MAPE, the combined model achieves the lowest prediction error, followed by the price-only model, while sentiment-only performs worst. The weak performance of sentiment-only is partly explained by data sparsity, where days without news were treated as neutral, reducing the usefulness of the sentiment signal.

II. RELATED WORKS

Early work on volatility prediction is dominated by econometric models such as ARCH and GARCH, which explicitly model time-varying variance in financial time series. While effective at capturing volatility clustering, these methods rely on statistical assumptions and may struggle with nonlinear market behavior. A common target in this setting is realized volatility, computed from recent returns. In this paper, we use a rolling-window realized volatility label

$$\sigma_t = \sqrt{\frac{1}{N} \sum_{i=1}^N r_{t-i}^2} \text{ where } r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Where P_t is the closing price at day t , r_t is the daily log return, and N is the rolling window window length

With the rise of deep learning, recurrent models, especially Long Short-Term Memory (LSTM) networks, have been widely applied to financial time-series forecasting. LSTMs can learn nonlinear temporal dependencies directly from historical price features, and multiple studies report that LSTM-based models can outperform traditional volatility baselines under suitable preprocessing and time-based evaluation.

More recently, researchers have explored augmenting price-based models with textual signals such as financial news sentiment. Domain-specific language models (e.g., FinBERT) allow extracting daily sentiment features, which may complement price information. However, the benefit of sentiment depends strongly on news coverage and aggregation choices. Motivated by this, our work compares price-only, sentiment-only, and fused price-sentiment models to assess when sentiment contributes to short-term volatility prediction.

III. RESEARCH QUESTIONS

RQ1. Does adding news sentiment improve volatility prediction beyond a price-only model?

RQ2. Is sentiment alone sufficient, or does it require price information to be competitive?

RQ3. Can low-dimensional embeddings (64-D) retain enough information to predict daily volatility well, and are they structured (e.g., PCA compressible)?

RQ4. How does missing news (days with no sentiment) affect the merged dataset and model behavior?

IV. Methodology: Numerical Volatility Feature Extraction

A. Overview

The numerical volatility modeling framework is designed to capture temporal price dynamics and estimate future stock volatility using historical market data. Instead of performing a simple regression on raw price movements, the model focuses on learning latent representations that encode short-term and long-term volatility patterns. To achieve this, we employ a sequential deep learning architecture based on **Long Short-Term Memory (LSTM)** networks, which are well-suited for modeling time-dependent financial data.

The numerical model is constructed as a modular component that can operate independently or serves as a numerical encoder, transforming raw OHLCV sequences into dense 64-dimensional embeddings for downstream multi-modal fusion, such as sentiment-based features, in a unified volatility prediction framework. Below, we outline the key components of the numerical modeling pipeline.

B. Data Source and Volatility Construction

The primary data source consists of daily historical price data (Open, High, Low, Close, Adjusted Close, and Volume).

- **Target Label (Ground Truth):** To represent market risk, we compute the **Realized Volatility** defined as the 21-day rolling standard deviation of daily log returns from the adjusted closing prices:

$$\text{Log Return}_t = \ln(P_t/P_{t-1})$$

$$\text{Volatility}_t = \text{std}(\text{Log Return}_{t-20}, \dots, \text{Log Return}_t)$$

- **Data Preprocessing:** We apply two separate MinMaxScaler to normalize all input features and target variables into a range of [0, 1], because they have vastly different magnitudes.

Realized Volatility ensures that the model learns from price fluctuations rather than absolute price levels, allowing it to

better capture market uncertainty and risk dynamics. In addition, Scaling is critical for LSTM convergence, ensuring that gradients remain stable during backpropagation. On top of that, all numerical data are aligned by trading date to preserve temporal consistency.

C. Sequential State Representation

Since volatility is inherently dependent on historical context, the model adopts a sliding window approach to construct sequential inputs. Each input sample consists of a fixed-length sequence of past numerical observations.

- Sequence Length: 60 trading days
- Input: Historical normalized price-based features
- Output: Volatility value at the next time step

This formulation enables the model to observe recent market behavior and capture short-term trends as well as longer-term volatility persistence.

D. LSTM-Based Numerical Encoder

The core of the numerical modeling framework is an LSTM-based neural network designed to encode temporal market information. The LSTM processes the input sequences and learns hidden representations that summarize historical volatility dynamics.

The architecture consists of:

- An input layer for sequential inputs
- Single-layer LSTM to model complex temporal dependencies creating a single 64-dim "context vector" for the entire window
- A fully connected dense layer for LSTM model training which is later **removed** to directly touch the embedding vector

The model is trained using the Mean Squared Error (MSE) loss function and optimized with the Adam optimizer. A chronological train-test split is applied to prevent future information leakage.

E. Numerical Embedding Extraction

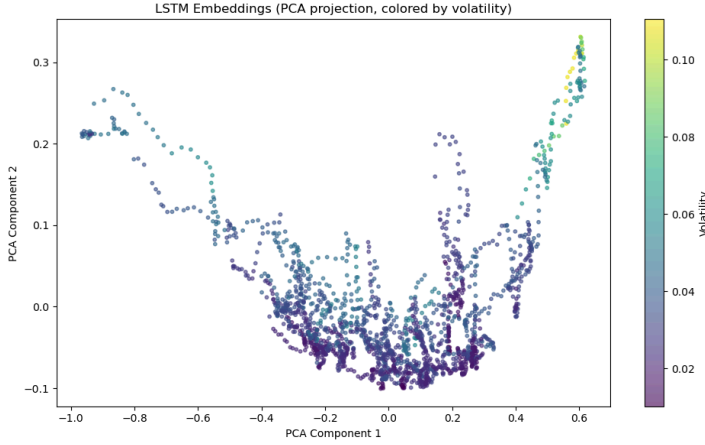
Rather than using the LSTM solely as a direct predictor, it performs as an encoder. Upon training convergence, the final regression layer (Dense) is removed, and the learned hidden representations are extracted as **numerical embeddings**. These embeddings encode high-level volatility-related features derived from historical price movements.

Each embedding corresponds to a specific time window and serves as a condensed representation of market behavior, making it suitable for downstream prediction tasks or multimodal fusion.

F. Validation and Visualizing

To validate the quality of the extracted embeddings, we employ **Principal Component Analysis (PCA)** as a diagnostic tool:

The 64-dimensional vectors are projected into a 2D space. By visualizing these points colored by their ground-truth volatility, we confirm that the embeddings form distinct clusters (e.g., high-volatility states are spatially separated from low-volatility states). This confirms that the LSTM has successfully learned to distinguish different market regimes.



G. Integration with Volatility Prediction System

The final numerical features embeddings are stored as structured dataset (`lstm_embeddings.csv`) where each trading date is associated with its corresponding 64-dimensional numerical fingerprint and used as inputs to the overall volatility prediction system. This modular design allows the numerical component to be evaluated independently or combined with additional data sources in a unified deep learning architecture.

By separating feature extraction from final prediction, the model enhances interpretability and flexibility while maintaining strong predictive performance.

V. RESULTS

TABLE I
TEST PERFORMANCE (REAL VOLATILITY UNITS) FROM `Model_3_DNN.ipynb`

Model	MSE	RMSE	MAE	MAPE	RMSP E
Price-only	0.000048	0.006953	0.005457	17.45%	21.93%
Sentiment-only	0.000055	0.007411	0.006033	20.08%	25.10%
Combined (Price + Sentiment)	0.000031	0.005608	0.004596	14.96%	18.93%

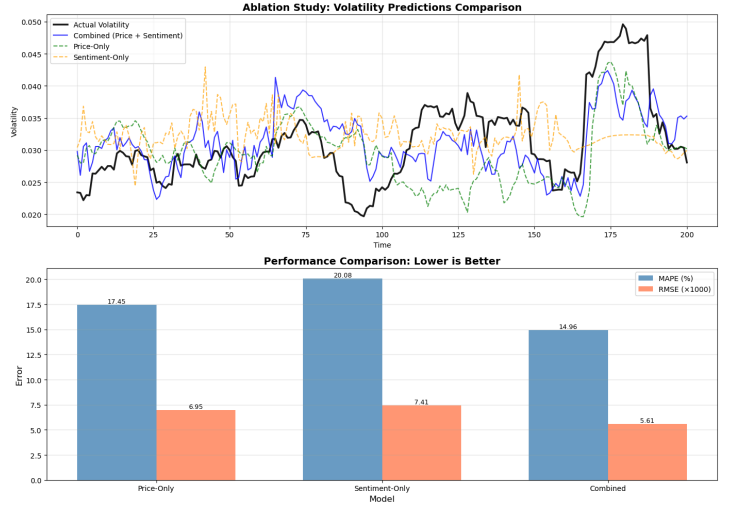


Fig. 1. Model Comparison for Volatility Forecasting: Price vs. Sentiment vs. Fusion (MAPE/RMSE)

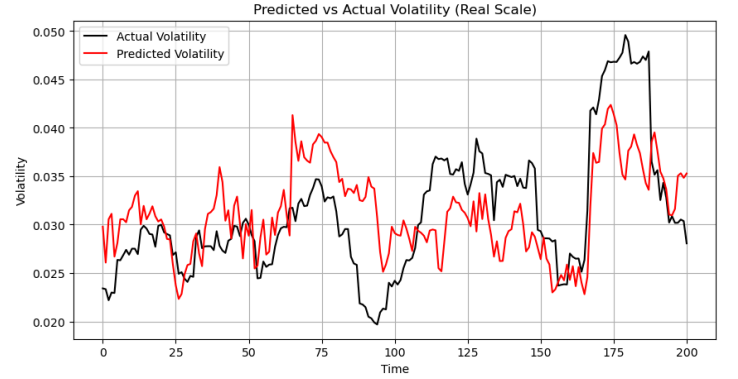


Fig.2. Predicted vs. actual realized volatility on the test set (real scale), using the combined model.

Answer (RQ1): Yes. Adding sentiment improves volatility prediction. In the test set, **MAPE drops from 17.45% → 14.96%** (−2.49 percentage points) and **RMSE improves from 0.006953 → 0.005608**.

Answer (RQ2): Sentiment-only is not sufficient here. It performs worst among the three models (MAPE **20.08%**), but the combined model shows that sentiment provides **complementary** information when fused with price embeddings.

Model 1 PCA result (from `Model_1_Price.ipynb`):

- PC1 explains **0.9077301**
- PC2 explains **0.07797362**
- Total (PC1+PC2) $\approx$ **0.98570372** ($\approx$ **98.6%**)

Answer (RQ3): The 64-D **price embedding** is highly compressible ($\approx$ 98.6% of variance in the first two PCs), suggesting a strongly **structured / low-dimensional** representation.

Merged dataset stats (from *Model_2_Sentiment.ipynb*):

- 2014 total trading days
- 772 days with actual sentiment
- 1242 days filled with 0.0 sentiment (no matched news days)

Also, embedding-based models use **SEQUENCE_LENGTH** = 60, so the embedding datasets contain 1954 samples (2014 – 60), as shown by the Train/Val/Test counts in *Model_3_DNN.ipynb*.

Answer (RQ4): Missing news is a major regime: 1242/2014 = 61.67% of trading days have no matched sentiment and are filled with 0.0. This means the model is trained on a dataset where “no news” is the **dominant case**, so the handling of missingness can materially affect learning and generalization. The notebooks report the missing-news counts at the merged-daily-table level (2014 rows), but they do not report the exact missing-news breakdown after the 60-day sequence offset (1954 samples). Future work should handle missingness more explicitly (e.g., add a missingness indicator/mask, learn a dedicated “no-news embedding,” or use better imputation than hard 0.0 fill).

VI. Discussion

A. The Complementary Nature of Price and Sentiment

Our experiments demonstrate that the combined model achieves superior performance with a MAPE of 14.96%. This validates the hypothesis that financial news sentiment contains information that is orthogonal to historical price data. While the Price-only model effectively captures the autoregressive nature of volatility, where past volatility predicts future volatility, it often struggles to anticipate sudden market shifts driven by external information shocks.

By integrating FinBERT-derived sentiment embeddings, the combined model gains access to the market narrative. The reduction in prediction error suggests that the neural network successfully learned to weigh the sentiment signal. It likely relies on price history for baseline continuity while utilizing sentiment impulses to adjust predictions during news-heavy periods. This confirms that price and text are not redundant but complementary. Price captures the market reaction, whereas sentiment often captures the catalyst.

B. Analyzing the Underperformance of Sentiment-Only Models

The Sentiment-only model performed the worst, yielding a MAPE of 20.08%. This result highlights a fundamental characteristic of financial time series. Volatility is heavily path-dependent. A model that ignores the price range or volume from the previous day lacks the context of the current market regime. It cannot determine

if the market is currently calm or chaotic.

Furthermore, sentiment is a sparse signal, while price is continuous. A day with negative sentiment might cause high volatility in a bear market but might be ignored in a strong bull market. Without price data to provide this context, the Sentiment-only model struggles to map sentiment scores directly to the magnitude of price fluctuations.

C. The Impact of Data Sparsity and Neutral Fill

A critical finding in our data analysis is that 61.67% of trading days lacked matching news and were imputed with a neutral sentiment score of 0.0. This sparsity poses a significant challenge. By treating “no news” as equivalent to “neutral news,” the model may be dampening the signal variance.

The zero-fill approach likely forces the LSTM to learn a bias toward lower volatility on days without news. This may not always reflect reality because markets can be volatile due to liquidity shocks or technical trading even in the absence of headlines. The fact that the combined model still outperformed the price-only model despite this noise suggests that the positive signal from the 38.33% of days with news was strong enough to overcome the dilution caused by the zero-filled days.

D. Latent Structure of Market Volatility

The PCA analysis of the price embeddings revealed that approximately 98.6% of the variance in the 64-dimensional LSTM hidden states could be explained by the first two principal components. This result is highly significant. It implies that while the raw OHLCV input data is high-dimensional and noisy, the underlying state of market volatility is actually low-dimensional.

The LSTM essentially learned to compress complex market dynamics into a simple 2D manifold. This likely represents the transition between distinct market regimes, such as stable low-volatility states versus high-volatility shock states. This compressibility suggests that future work could achieve similar performance with smaller model architectures or by explicitly modeling these regimes rather than relying on deep, high-dimensional vectors.

E. Limitations

While effective, our approach relies on a rolling-window label of 21-day realized volatility. This introduces a lag in the ground truth. Additionally, the daily aggregation of sentiment scores averages out intra-day nuances. A day with highly positive and highly negative news might average to zero, which mimics a day with no news. Finally, the current method of handling missing news through hard 0.0 imputation is a naive approach. Future iterations could employ attention mechanisms to allow the model to dynamically ignore the sentiment input when news is absent rather than

interpreting it as neutrality.

VII. CONCLUSION

RQ1: Yes, fusing sentiment with price improves forecasting (**MAPE 14.96% vs 17.45%**).

RQ2: Sentiment alone is weaker (**MAPE 20.08%**) but contributes useful complementary signal in fusion.

RQ3: Price embeddings are highly PCA-compressible (**≈98.6%** variance in first two PCs), indicating structured representations.

RQ4: Missing news is common (**61.67%** of days are filled with 0.0 sentiment). Missingness handling likely matters; the current notebooks use neutral fill and do not fully analyze post-sequence missingness effects.

VIII. AUTHOR CONTRIBUTION STATEMENTS

- **J. Malvin** (16.65%): study research, methodology design, result analysis, evaluation metrics, data curation
- **C. Richard** (16.65%): methodology design, study research
- **Y. HU** (16.65%): methodology design, study research
- **T. Tsai** (16.65%): video presentation, research questions, results, conclusion
- **H. Hsu** (16.65%): Discussion, limitations, result analysis
- **Waylen T. Collin** (16.65%): methodology design, study research

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